

Demonstration of Ultra-high Link Budget, Low Cost/Latency Distributed AI Scale-across Network enabled by C-band 3.2-Tb/s/lane IM-DD Transmission over 20-km AR-HCF

Jingchuan Wang⁽¹⁾, Yibin Li⁽¹⁾, Peng Li⁽²⁾, Lei Zhang⁽²⁾, Jie Luo⁽²⁾, Kangping Zhong⁽¹⁾, Changyuan Yu⁽¹⁾, Alan Pak Tao Lau⁽¹⁾, Chao Lu⁽¹⁾,

⁽¹⁾ Photonics Research Institute, Department of Electrical and Electronic Engineering, The Hong Kong Polytechnic University, China, yibinxm.li@connect.polyu.hk

⁽²⁾ State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Yangtze Optical Fiber and Cable Joint Stock Limited Company, Wuhan 430074, China.

Abstract We demonstrate a C-band $32\lambda \times 50$ -GBaud PAM4 transmission over 20-km HCF. We achieve a record link budget over 50-dB, showing that HCF-enabled IM-DD technology is strong candidate for AI scale-across scenarios requiring low-latency, low-cost, and zero packet loss. ©2026 The Author(s)

Introduction

As AI training workloads grow, while computing capacity and power within individual campuses remain limited, distributed AI training across campuses or even regions is becoming inevitable [1]. Although optical link capacity can be scaled through dimensional multiplexing, distributed AI training imposes three additional requirements: zero packet loss, cost- and power-efficient operation, and large-scale synchronized training [2]. Accordingly, the underlying fiber networks face major challenges in meeting these demands: 1) Achieving near-zero packet loss usually requires stronger forward error correction (FEC) that can operate at higher pre-FEC bit-error ratios (BERs). This often raises the minimum required received optical power (ROP), thereby reducing the available power budget. 2) Any increase in the cost or power consumption of an optical link is amplified by the enormous aggregate bandwidth demand of distributed AI systems. 3) Low latency is essential for large-scale synchronized training, where many computational units must frequently exchange and synchronize parameters and gradients.

However, simultaneously meeting all three requirements remains unsolved. Coherent technology provides better receiver sensitivity and thus a larger power budget; however, these advantages come at the cost of greater hardware complexity and digital signal processing (DSP) latency, as well as higher cost and power consumption [3]. In contrast, intensity-modulation/direct-detection (IM-DD) technology generally offers lower cost and lower DSP latency, but suffers from a smaller power budget and poorer signal integrity [4]. Moreover, link latency is fundamentally limited by propagation delay, which constrains the maximum deployable radius for large-scale synchronized AI training.

The emergence of hollow-core fiber (HCF) [5] makes IM-DD-based dense wavelength-division multiplexing (DWDM) networks a promising solution to this dilemma, owing to several key advantages over standard single-mode fiber (SSMF): 1)

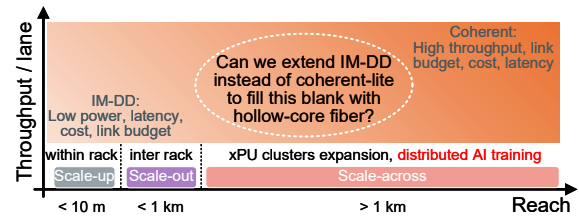


Fig. 1: Extend IM-DD to 2-40 km scale-across via HCF.

The nonlinear coefficient of HCF is about 30 dB lower than that of SSMF [6], enabling significantly higher launch power and improving the achievable power budget. 2) In the C-band, the chromatic dispersion (CD) of HCF is typically around 3 ps/nm/km, substantially mitigating CD-induced power fading in high-baud-rate IM-DD systems [7]. 3) Light propagates about 1.5× faster in HCF than in SSMF. Consequently, under a given latency constraint, the maximum deployable radius for distributed AI training can be extended by up to 2.25×. These advantages make HCF a strong candidate for extending IM-DD technology to 2-40 km distributed AI training, as shown in Fig. 1.

Motivated by these findings, we experimentally demonstrate a C-band $32\lambda \times 50$ -GBaud PAM4 transmission over a 20-km fiber link, achieving a record 50-dB link budget. Unlike many previous studies [8–10], we provide a quantitative comparison of HCF and SSMF under both single-wavelength and DWDM operation. Our results show that, under the combined effects of CD, Kerr nonlinearity, stimulated Brillouin scattering (SBS), Rayleigh scattering, and laser instability, a 20-km SSMF link cannot sustain launch powers above 11 dBm, limiting the achievable link budget. In contrast, HCF effectively overcomes these limitations, increasing LP to over 30 dBm in a DWDM system.

Practical Constraints and Experimental Setup

It is well known that, in high-baud-rate IM-DD systems, achievable transmission distance is limited by CD-induced power fading [11]. For example, in a 20-km standard single SSMF link with a typical CD coefficient of 16.7 ps/nm/km, the first fading

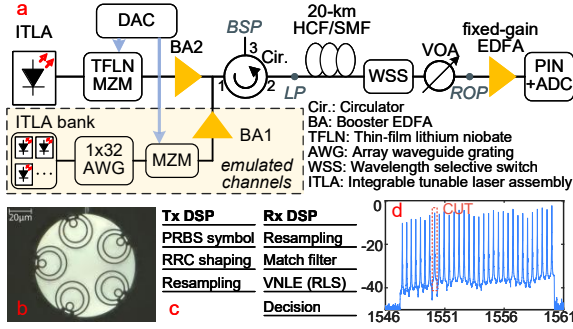


Fig. 2: (a) IM-DD DWDM setup. (b) Cross-section of support-tube HCF. (c) Tx/Rx DSP. (d) DWDM spectrum.

null appears at about 13.7 GHz. Thus, for IM-DD signals above 28-GBaud, at least one spectral notch lies within the signal bandwidth. More importantly, under the combined effects of CD, Rayleigh backscattering, multipath interference (MPI), and laser frequency instability, the fading positions can fast fluctuate over time, as shown in a later section.

Conventionally, to reduce accumulated CD, DWDM IM-DD data are transmitted in the O-band. However, this introduces a more severe limitation: fiber nonlinearity. Because CD is minimal, dispersion-induced walk-off between channels is reduced, enhancing nonlinear interactions governed by the Kerr effect [12]. As a result, the nonlinear penalty becomes larger, restricting the allowable launch power and achievable link budget. In addition, stimulated Brillouin scattering (SBS) further limits the maximum launch power. Consequently, O-band DWDM IM-DD systems struggle to provide the large link budgets required for applications such as distributed AI training.

Fortunately, HCF offers a promising solution to mitigate the practical constraints of deploying IM-DD DWDM technology in scale-across scenarios. To verify its feasibility and benchmark it against SSMF-based systems, a C-band DWDM IM-DD experimental setup was built, as shown in Fig. 2a. A low-cost laser at 1550.12 nm was used as the channel under test (CUT). The IM/DD data stream was generated by a 120-GSa/s digital-to-analog converter (DAC) (Keysight M8194A) and externally modulated by a 70-GHz TFLN MZM. The modulated signal was then amplified by a booster EDFA (Amonics BA-33) to vary the launch power. To emulate DWDM operation, 31 additional lasers with 50-GHz channel spacing were combined with the channel under test (CUT) using a 1×32 AWG. The multiplexed channels were then modulated by a MZM to load all wavelengths with data at the same baud rate as the CUT. After passing through an optical circulator, the aggregated 32-channel DWDM signal was launched into a 20-km fiber. Two fiber types were evaluated: G.652D SSMF and a support-tube HCF (ST-HCF) manufactured by YOFC [13], whose cross-section is shown in Fig. 2b. At the receiver, the CUT was first selected by a WSS with bandwidth matched to the baud rate, and then passed through a variable optical attenuator (VOA) to control the ROP. The signal

was subsequently amplified by a fixed-gain EDFA to compensate for the sensitivity penalty of the PIN photodiode. The electrical output of the PIN was then digitized by a 256-GSa/s analog-to-digital converter (ADC) for offline processing.

The DSP flows at both the transmitter and receiver are shown in Fig. 2c. At the transmitter, a root-raised-cosine (RRC) filter with a roll-off factor of 0.01 was used for pulse shaping. At the receiver, a Volterra nonlinear equalizer (VNLE) with 101 linear taps and 15 nonlinear taps was employed, with coefficients updated using the recursive least squares (RLS) algorithm [14]. Fig. 2d shows the optical spectrum of the 32-channel DWDM signal. It should be noted that the spectrum nonuniformity is < 5 dB, mainly due to the gain tilt of the EDFA.

Results and Discussions

A. Single wavelength (C34) analysis

We first disable other wavelengths to assess single-wavelength performance under three configurations: 20-km HCF, 20-km SMF, and optical back-to-back (OB2B).

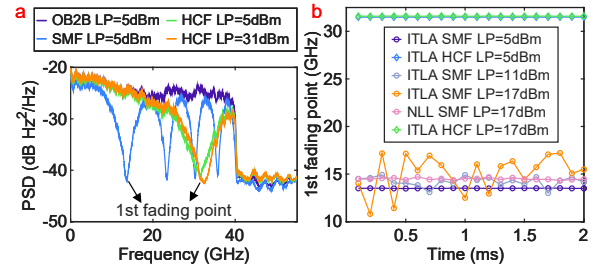


Fig. 3: (a) Received PSD of 80-GBaud PAM4. (b) The shift of first fading point with time, LP and fiber.

An 80-GBaud PAM4 signal is first transmitted to examine CD and fiber nonlinearity. As shown in Fig. 3a, the received power spectral density (PSD) at a launch power (LP) of 5 dBm shows that the first fading notch occurs at 13.5 GHz for 20-km SMF and 31.5 GHz for 20-km HCF. This indicates that the CD coefficient of ST-HCF is about 3 ps/nm/km in the C-band. Moreover, when the LP in HCF is increased to 31 dBm, the fading position show negligible shift, indicating an extremely low self phase modulation (SPM) in the HCF link [15].

We next explain why, from a CD perspective, a 20-km SMF link cannot sustain high LP. As shown in Fig. 3b, when an ITLA is used, the fading frequency gradually shifts to higher frequencies as the LP increases from 5 dBm to 11 dBm. This behavior is attributed to the positive chirp induced by SPM. However, once the LP exceeds 11 dBm, the fading frequency begins to fluctuate abruptly, with an oscillation amplitude of several GHz, as observed at an LP of 17 dBm. This apparent drift is mainly caused by laser frequency instability, which dynamically sweeps the parasitic interference pattern arising from Rayleigh backscattering and MPI, thereby introducing a rapidly varying effective chirp. When the ITLA is replaced with a frequency-stable narrow-linewidth laser (NLL), the interference pattern remains nearly stationary, and the fading fre-

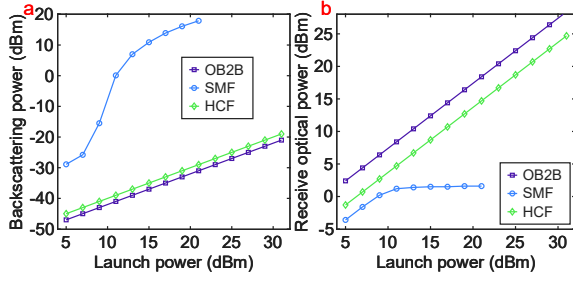


Fig. 4: (a) BSP vs LP. (b) ROP vs LP.

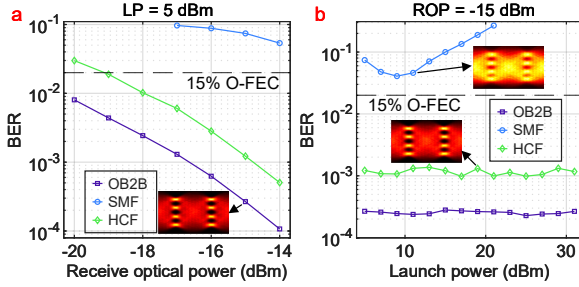


Fig. 5: BER vs (a) ROP and (b) LP of 50-Gbaud PAM4.

quency stabilizes at about 14.2 GHz even at an LP of 17 dBm. In practical IM/DD systems, however, NLLs are prohibitively expensive for low-cost deployment. Consequently, increasing the LP of a 20-km SSMF link beyond 11 dBm is highly challenging. In contrast, HCF is largely immune to the fast-varying chirp induced by parasitic interference because of its negligible Rayleigh scattering, enabling stable high-power operation.

Another factor limiting LP of the SMF beyond 11 dBm is SBS. As shown in Fig.4a, for the 20-km SMF link, BSP rises sharply once the LP exceeds 11 dBm. In contrast, for both the HCF and OB2B configurations, the BSP scales linearly with LP. This indicates that, in SSMF, the backscattered signal is dominated by SBS, whereas in the OB2B and HCF cases it is mainly governed by linear reflections. The impact of SBS is further confirmed in Fig.4b. HCF exhibits lower loss than SSMF and its ROP increases linearly with LP, consistent with the OB2B case. By comparison, in the SMF link the ROP saturates once the LP exceeds 11 dBm, confirming that SBS clamps the transmitted power.

We then evaluate the communication performance of a single-wavelength 50-Gbaud PAM4 signal to align with the DWDM scenario. First, the impact of ROP on BER was investigated at a fixed LP of 5 dBm. As shown in Fig.5a, when the ROP exceeds -19 dBm, the BER of the HCF system remains below the 15% O-FEC threshold [16]. In contrast, due to pronounced CD, the BER of the SMF system remains above the threshold under the same conditions. Fig.5b further shows that the BER of the HCF system exhibits negligible degradation as the LP is increased to 31 dBm, demonstrating strong tolerance to high launch power. By comparison, the BER of the SMF system reaches an optimum at an LP of about 9 dBm, aligned with the earlier analysis of CD-induced fading, Kerr nonlinearity, and SBS effects.

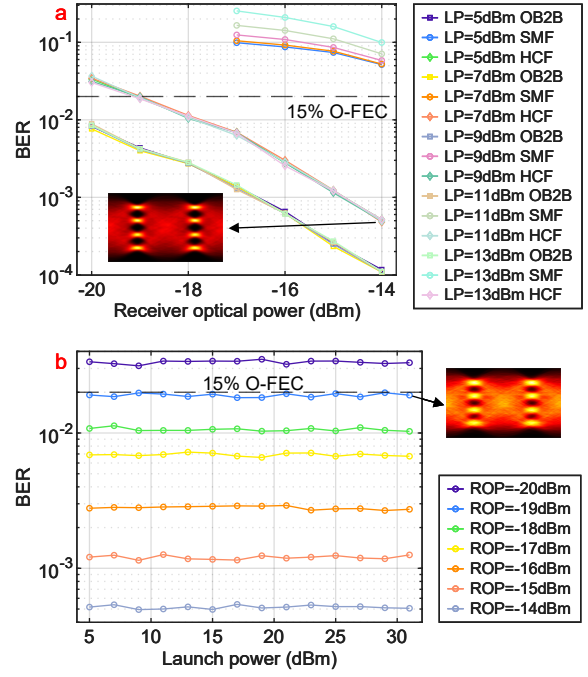


Fig. 6: BER vs (a) ROP and (b) HCF LP of DWDM case.

B. $32 \lambda \times 50\text{GHz}$ spacing DWDM transmission

Next, we evaluate the system performance of SMF and HCF under DWDM operation. As shown in Fig. 6a, the LP denotes the average power per wavelength. Due to the gain limitations of the EDFA, the additional 31 wavelengths could only be amplified to a maximum average LP of 13 dBm. Under both OB2B and HCF conditions, the measured BER is comparable to that of the single-channel case, indicating that the additional noise introduced by DWDM multiplexing is not the dominant impairment. In contrast, the SMF system achieves its optimal performance at an LP of 5 dBm, as higher launch powers introduce additional nonlinear distortions, including four-wave mixing (FWM) and cross-phase modulation (XPM).

Because the EDFA could not simultaneously amplify all wavelengths to 31 dBm, we further investigated a hybrid scenario in which CUT was boosted to 31 dBm, while the remaining channels were maintained at 13 dBm. As shown in Fig. 6b, the BER of the HCF system remains nearly unchanged as the CUT LP increases to 31 dBm, demonstrating strong robustness to high-power operation. At a ROP of -19 dBm, a link budget of 50 dB was achieved while simultaneously transmitting a 3.2-Tb/s line rate under the 15% O-FEC threshold. To the best of our knowledge, this is the highest link budget in DWDM IM-DD transmission.

Conclusion

We experimentally demonstrated a C-band $32\lambda \times 50\text{-Gbaud}$ PAM4 transmission over 20-km HCF, achieving a 3.2-Tb/s line rate and a record 50-dB link budget. Compared with SSMF, HCF enables IM-DD system with stable launch power and lower latency, making it a compelling low-cost solution for distributed AI scale-across interconnects.

Acknowledgements

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